

Classical ILC/DRP Design from an Algebraic (Matrix-Fraction) Perspective – Part 1: Framework

Kevin L. Moore

Department of Electrical Engineering
Colorado School of Mines
Golden, Colorado, USA

**Pre-Congress Workshop WS-13 – Learning from Data: Principles, Methods,
and Emerging Ideas & Applications in Iterative Learning Control**

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Dedicated to the Memory of YangQuan Chen



A Simple Idea ...

...old enough to be one of those Latin phrases that's engraved into stone: *Repetitio mater studiorum est*, or **“Repetition is the mother of all learning”**



www.brainscape.com/blog/2011/05/repetition-is-the-mother-of-all-learning

Control Paradigms Involving Repetition

- **Iterative Learning Control (ILC)**
 - For systems that seek to execute the same trajectory over-and-over from the same starting point each time
 - In this case the control input is adapted from one trial to the next



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- **Repetitive Control (RC)**
 - For systems subject to periodic references or disturbances
 - For systems with periodic parameter variations
- **(Discrete) Repetitive Processes (DRP)**
 - Same operational scenario as ILC, but ...
 - System has inherent trial-to-trial memory



In This Two-Part Talk ...

- **Part 0: Brief Overview of two of these paradigms:**
 - ILC
 - DRP
- **Part 1: Analysis and design framework for such systems**
 - Lifted models
 - The “supervector” framework of ILC
 - The update law using supervector notation
 - The ILC design problem
 - A lifting approach to DRPs
 - The w -transform and operator perspective
 - Generalization and closed-loop control of DRPs
 - Complete framework
- **Part 2: Design for discrete ILC/DRP systems**



Iterative Learning Control (ILC)

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 - Finite time or spatial interval of operation
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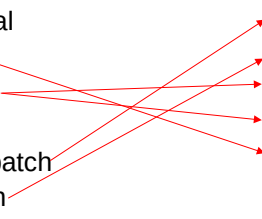


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 - Iterative
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 - Periodic
 - Batch-to-batch
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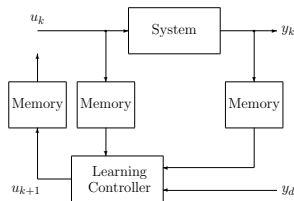


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 - ...
 - Such systems arise in a variety of applications, including:
 - Process control
 - Semiconductor processing
 - Robotic motion control
 - Hard disk drive servo control
 - Machining and manufacturing operations
 - ...
- 

ILC Algorithms

- Standard iterative learning control scheme:



- Goal:** Find a learning control algorithm

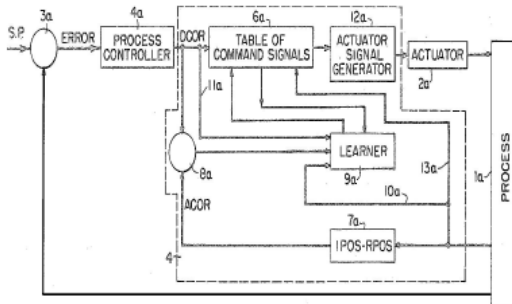
$$u_{k+1}(t) = f_L(\text{previous information})$$

so that for all $t \in [0, t_f]$

$$\lim_{k \rightarrow \infty} y_k(t) = y_d(t)$$

On the History and Accomplishments of ILC

- United States Patent 3,555,252 – “Learning Control of Actuators in Control Systems,” filed 1967, awarded 1971, learned characteristics of actuators and used this knowledge to correct command signals.



- “Comments on United States Patent 3,555,252 – Learning Control of Actuators in Control Systems,” Yangquan Chen and Kevin L. Moore, in *Proceedings of the 2000 International Conference on Automation, Robotics, and Control*, Singapore, December 2000.

First ILC paper- in English (1984)

S. Arimoto, S. Kawamura, and F. Miyazaki, "Bettering operation of robots by learning," *J. of Robotic Systems*, 1(2):123-140, 1984.

Bettering Operation of Robots by Learning

Suguru Arimoto, Sadao Kawamura, and Fumio Miyazaki
Faculty of Engineering Science, Osaka University, Toyonaka, Osaka,
560 Japan
Received January 26, 1984; accepted March 12, 1984

This article proposes a betterment process for the operation of a mechanical robot in a sense that it better the next operation of a robot by using the previous operation's data. The process has an adaptive learning structure such that the $(k+1)$ th input to joint actuators consists of the k th input plus an error increment composed of the derivative difference between the k th motion trajectory and the given desired motion trajectory. The convergence of the process to the desired motion trajectory is assured under some reasonable conditions. Numerical results by computer simulation are presented to show the effectiveness of the proposed learning scheme.

紙面の所載のデータを用いて、機械的ロボットシステムの動作を改善するプロセスを提示する。このプロセスは、前回の動作データを用いて、次の動作に改善を加える。改善は、前回の動作データと、目標動作データの差分の導関数を用いて行われる。このプロセスが、目標動作に収束することを、いくつかの合理的な条件下で保証する。数値的シミュレーションの結果を示す。

1. INTRODUCTION

It is human to make mistakes, but it is also human to learn much from experience. Athletes have improved their form of body motion by learning through repeated training, and skilled hands have mastered the operation of machines or plants by acquiring skill in practice and gaining knowledge from experience. Upon reflection, can machines or robots learn autonomously (without the help of human beings) from measurement data of previous operations and make better their performance of future operations? Is it possible to think of a way to implement such a learning ability in the automatic operation of dynamic systems? If there is a way, it must be applicable to allowing autonomy and intelligence to industrial robots.

Motivated by this consideration, we propose a practical approach to the problem of bettering the present operation of mechanical robots by using the data of

Journal of Robotic Systems, 1(2), 123-140 (1984)
© 1984 by John Wiley & Sons, Inc. CCC 0891-2223/84/020123-18\$04.00

Professor Arimoto's Early Contributions

- Professor Arimoto's Six Postulates of ILC:

- **P1.** Every trial (pass, cycle, batch, iteration, repetition) ends in a fixed time of duration $T > 0$.
- **P2.** A desired output $y_d(t)$ is given *a priori* over $[0, T]$.
- **P3.** Repetition of the initial setting is satisfied, that is, the initial state $x_k(0)$ of the objective system can be set the same at the beginning of each iteration: $x_k(0) = x^0$, for $k = 1, 2, \dots$.
- **P4.** Invariance of the system dynamics is ensured throughout these repeated iterations.
- **P5.** Every output $y_k(t)$ can be measured and therefore the tracking error signal, $e_k(t) = y_d(t) - y_k(t)$, can be utilized in the construction of the next input $u_{k+1}(t)$.
- **P6.** The system dynamics are invertible, that is, for a given desired output $y_d(t)$ with a piecewise continuous derivative, there exists a unique input $u_d(t)$ that drives the system to produce the output $y_d(t)$.

- Professor Arimoto proposed a learning control algorithm of the form:

$$u_{k+1}(t) = u_k(t) + \Gamma \dot{e}_k(t)$$

Convergence is assured if $\|I - CB\Gamma\|_i < 1$.

- Professor Arimoto also considered more general algorithms of the form:

$$u_{k+1} = u_k + \Phi e_k + \Gamma \dot{e}_k + \Psi \int e_k dt$$

Other Learning Control Algorithms

- Numerous studies have considered a variation on the theme of the basic Arimoto algorithm, in both continuous and discrete-time, using **contraction mapping-based analysis and design**:

$$u_{k+1}(t) = u_k(t) + \Gamma(s)e_k(t)$$

- Various researchers have used **gradient methods** to optimize the gain G_k in:

$$u_{k+1}(t) = u_k(t) + G_k e_k(t+1)$$

- **Norm-optimal ILC** works in the state space. For the plant:

$$x_{k+1}(t+1) = Ax_{k+1} + Bu_{k+1}$$

find u_{k+1} to minimize

$$J = \|u_{k+1} - u_k\|^2 + \|e_{k+1}\|^2$$

- It is also useful for analysis and design use **frequency domain-based learning control**, for example:

$$U_{k+1}(s) = L(s)[U_k(s) + aE_k(s)]$$

- A significant amount of work has also been done for nonlinear ILC, leading to the so-called **energy function approach**, using what are called control Lyapunov functions.
- Lots of work using **fuzzy-logic and artificial neural networks**.



Operator-Theoretic Convergence Condition

- **Theorem** (Moore, *et al.*, 1988): For the plant $y_k = T_s u_k$, the linear time-invariant learning control algorithm

$$u_{k+1} = T_u u_k + T_e (y_d - y_k)$$

converges to a fixed point $u^*(t)$ given by

$$u^*(t) = (I - T_u + T_e T_s)^{-1} T_e y_d(t)$$

with a final error

$$e^*(t) = \lim_{k \rightarrow \infty} (y_k - y_d) = (I - T_s(I - T_u + T_e T_s)^{-1} T_e) y_d(t)$$

defined on the interval (t_0, t_f) if $\|T_u - T_e T_s\|_i < 1$



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- **Observation:**
 - If $T_u = I$ then $\|e^*(t)\| = 0$ for all $t \in [t_0, t_f]$.
 - Otherwise the error will be non-zero.

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- **Observation:**
 - If $T_u = I$ then $\|e^*(t)\| = 0$ for all $t \in [t_0, t_f]$.
 - Otherwise the error will be non-zero.
- **Conclusion:** The essential effect of ILC is to *produce the output of the best possible inverse of the system in the direction of y_d .*



For Further Reading ...

- K. L. Moore, M. Dahleh, and S. P. Bhattacharyya, "Iterative learning control: a survey and new results," *J. of Robotic Systems*, 9(5):563–594, 1992.
- K.L. Moore, "Iterative learning control - an expository overview," *Applied & Computational Controls, Signal Processing, and Circuits*, 1(1):151–241, 1999.
- D.A. Bristow, M. Tharayil, and A. G. Alleyne, "A Survey of Iterative Learning Control," *IEEE Control Systems Magazine*, 26(3): 96–114, 2006.
- H. S. Ahn, Y. Q. Chen, and K. L. Moore, "Iterative learning control: brief survey and categorization 1998 – 2004," *IEEE Trans. on Systems, Man, and Cybernetics, Part-C*, Vol. 37, 2007.
- Y. Wang, F. Gao, F. J. Doyle III, "Survey on Iterative Learning Control, Repetitive Control, and Run-to-Run Control," *Journal of Process Control*, Vol. 19, No. 10, pp. 1589-1600, 2009.
- Jian-Xin Xu, "A survey on iterative learning control for nonlinear systems," *International Journal of Control*, Vol. 84, No. 7, pp. 1275-1294, 2011.



ILC and Repetitive Control

- Much work has shown that RC can be viewed as a limiting case of ILC as the trial length goes to infinity
- For more see
 - “Unified Analysis of Iterative Learning and Repetitive Controllers in Trial Domain,” Goele Pipeleers and Kevin L. Moore, *IEEE Transactions on Automatic Control*, Vol. 59, No. 4, pp. 953-965, April 2014.

(Discrete) Repetitive Processes (DRP)

- There are a number of repetitive processes where the results of the previous iteration affect the plant seen by the controller in the next iteration
 - Material removal or application processes are prototypical
 - **Examples:**
 - Diffusion process
 - Digging
 - Multi-pass welding
 - Soil compaction
- Rather than ILC, these processes better fit a class of systems called **Repetitive Processes**

Typical DRP processes

- Seminal work was in the area of long-wall coal-mining (Edwards, et al)
- Metal rolling
 - Both of these are described in detail in *Control Systems Theory and Applications for Linear Repetitive Processes*, Rogers, Galkowski, and Owens, Lecture notes in Control and Information Sciences, no. 349, Springer-Verlag, Berlin Heidelberg, 2007.
- Stroke rehab/adaptive physical therapy
- Additive manufacturing



Modeling Repetitive Processes

Normal Discrete-Time Linear Model (no repetition)

$$x(p+1) = Ax(p) + Bu(p)$$

$$y(p) = Cx(p) + Du(p) \quad 0 \leq p \leq \alpha$$

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k is trial or iteration

Iterative Learning Control

Normal Discrete-Time ILC

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$$u_{k+1}(p) = u_k(p) + \Gamma(y_d(p+1) - y_k(p+1))$$

Iterative Learning Control

Normal Discrete-Time ILC

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ILC adds this control law to normal linear model

- **Modifies control not plant**
- **Adds memory to the control law**

Discrete Repetitive Processes

Discrete Repetitive Processes (DRP)

$$\begin{aligned}x_{k+1}(p+1) &= Ax_{k+1}(p) + Bu_{k+1}(p) + B_0 y_k(p) \\ y_{k+1}(p) &= Cx_{k+1}(p) + Du_{k+1}(p) + D_0 y_k(p) \quad 0 \leq p \leq \alpha\end{aligned}$$

Boundary conditions :

$$x_{k+1}(0) = d_{k+1}$$

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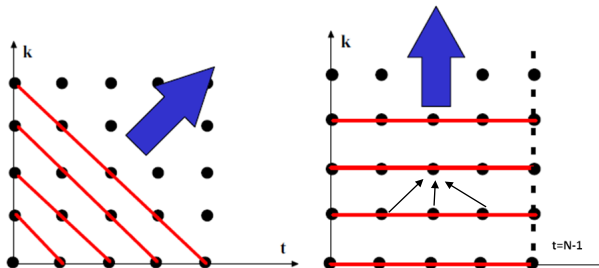
Control :

$$u_{k+1}(p) = f(y_d(p) - y_{k+1}(p))$$

Can then add “ILC-like” control

Repetitive Processes are 2-D Systems

- Repetitive processes propagate information in two independent directions where the duration of one of these is finite
- Repetitive Processes are a special type of 2-D systems (image concept due to Eric Rogers)



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 - **Complete framework**
- Part 2: Design for discrete ILC/DRP systems



The “Supervector” Framework of ILC -1

- Consider an SISO, LTI discrete-time plant with relative degree m

$$\begin{aligned} Y(z) &= H(z)U(z) \\ &= (h_m z^{-m} + h_{m+1} z^{-(m+1)} + h_{m+2} z^{-(m+2)} + \dots)U(z) \end{aligned}$$

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- By “lifting” along the time axis, for each trial k define

$$\begin{aligned} U_k &= [u_k(0), u_k(1), \dots, u_k(N-1)]^T \\ Y_k &= [y_k(m), y_k(m+1), \dots, y_k(m+N-1)]^T \\ Y_d &= [y_d(m), y_d(m+2), \dots, y_d(m+N-1)]^T \end{aligned}$$

The “Supervector” Framework of ILC -2

- Thus the linear plant can be described by $Y_k = H_p U_k$ where (for $m=1$)

$$H_p = \begin{bmatrix} h_1 & 0 & 0 & \dots & 0 \\ h_2 & h_1 & 0 & \dots & 0 \\ h_3 & h_2 & h_1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ h_N & h_{N-1} & h_{N-2} & \dots & h_1 \end{bmatrix}$$

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- The lower triangular matrix H_p is formed using the system's Markov parameters
- Notice the non-causal shift ahead in forming the vectors U_k and Y_k

The “Supervector” Framework of ILC -3

- The lifting operation over a finite interval allows us to represent our dynamical system in R^1 as a static system in R^N

The “Supervector” Framework of ILC -3

- The lifting operation over a finite interval allows us to represent our dynamical system in R^1 as a static system in R^N
- We can extend this to
 - Time-varying systems
 - The ILC update law
 - Discrete repetitive processes (DRP)
 - Repetitive control (RC)
 - Affine nonlinear systems

The “Supervector” Framework of ILC -4

- For the linear, time-varying case, suppose we have the plant given by

$$x_k(t+1) = A(t)x_k(t) + B(t)u_k(t)$$

$$y_k(t) = C(t)x_k(t) + D(t)u_k(t)$$

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$$H_p = \begin{bmatrix} h_{1,0} & 0 & 0 & \dots & 0 \\ h_{2,0} & h_{1,1} & 0 & \dots & 0 \\ h_{3,0} & h_{2,1} & h_{1,2} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ h_{N,0} & h_{N-1,1} & h_{N-2,2} & \dots & h_{1,N-1} \end{bmatrix}$$

The Update Law Using Supervector Notation -1

- Suppose we have a simple “Arimoto-style” ILC update equation with a constant gain γ , described in our R^1 representation as

$$u_{k+1}(t) = u_k(t) + \gamma e_k(t+1)$$

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$$\Gamma = \begin{bmatrix} \gamma & 0 & 0 & \dots & 0 \\ 0 & \gamma & 0 & \dots & 0 \\ 0 & 0 & \gamma & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \gamma \end{bmatrix}$$

The Update Law Using Supervector Notation -2

- Suppose we filter with a causal LTI filter of relative degree m during the ILC update, written in our R^1 representation as

$$u_{k+1}(t) = u_k(t) + L(z)e_k(t+1)$$

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$$L = \begin{bmatrix} L_m & 0 & 0 & \dots & 0 \\ L_{m+1} & L_m & 0 & \dots & 0 \\ L_{m+2} & L_{m+1} & L_m & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ L_{m+N-1} & L_{m+N-2} & L_{m+N-3} & \dots & L_m \end{bmatrix}$$

The Update Law Using Supervector Notation -3

- The supervector notation can also be applied to other ILC update schemes



The Update Law Using Supervector Notation -3

- The supervector notation can also be applied to other ILC update schemes
 - E.g., the Q -filter often introduced for stability (along the iteration domain) has the R^1 representation

$$u_{k+1}(t) = Q(z)(u_k(t) + L(z)e_k(t+1))$$

The Update Law Using Supervector Notation -3

- The supervector notation can also be applied to other ILC update schemes
 - E.g., the Q -filter often introduced for stability (along the iteration domain) has the R^1 representation

$$u_{k+1}(t) = Q(z)(u_k(t) + L(z)e_k(t+1))$$

- The equivalent R^N representation is

$$U_{k+1} = Q(U_k + LE_k)$$

where Q is a Toeplitz matrix formed using the Markov parameters of the filter $Q(z)$

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- We may also consider time-varying and noncausal filters in the ILC update law

The Update Law Using Supervector Notation -4

- A causal (in time), time-varying filter in the ILC update law might look like, for example

$$L = \begin{bmatrix} n_{1,0} & 0 & 0 & \dots & 0 \\ n_{2,0} & n_{1,1} & 0 & \dots & 0 \\ n_{3,0} & n_{2,1} & n_{1,2} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ n_{N,0} & n_{N-1,1} & n_{N-2,2} & \dots & n_{1,N-1} \end{bmatrix}$$

The Update Law Using Supervector Notation -5

- A non-causal (in time), time-invariant averaging filter in the ILC update law might look like, for example

$$L = \begin{bmatrix} K & K & 0 & 0 & \dots & 0 & 0 & 0 \\ 0 & K & K & 0 & \dots & 0 & 0 & 0 \\ 0 & 0 & K & K & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \dots & K & K & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 & K & K \\ 0 & 0 & 0 & 0 & \dots & 0 & 0 & K \end{bmatrix}$$

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 - L can also be fully populated

The ILC Design Problem -2

- Motivated by these comments, we will refer to the “causal” and “non-causal” elements of a general matrix Γ as follows

$$\Gamma = \begin{bmatrix} \gamma_{11} & \gamma_{12} & \gamma_{13} & \dots & \gamma_{1N} \\ \gamma_{21} & \gamma_{22} & \gamma_{23} & \text{noncausal} & \gamma_{2N} \\ \gamma_{31} & \gamma_{32} & \gamma_{33} & \dots & \gamma_{3N} \\ \vdots & \text{causal} & \vdots & \ddots & \vdots \\ \gamma_{N1} & \gamma_{N2} & \gamma_{N3} & \dots & \gamma_{NN} \end{bmatrix}$$

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The diagonal elements of Γ are referred to as “Arimoto” gains

A Lifting Approach to DRPs -1

- Consider a SISO, LTI, DRP defined on $0 \leq t \leq N$ and $k \in [0, \infty)$

$$x_{k+1}(t+1) = Ax_{k+1}(t) + Bu_{k+1}(t) + B_0y_k(t)$$

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where

$$H^u(z) = C(zI - A)^{-1}B + D$$

$$H^y(z) = C(zI - A)^{-1}B_0 + D_0$$

A Lifting Approach to DRPs -2

- We convert the 2-D SISO DRP into a 1-D MIMO system by using “supervectors”

$$U_k = (u_k(0), u_k(1), \dots, u_k(N-1))^T$$

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- This gives the lifted representation

$$Y_{k+1} = H^u U_{k+1} + H^y Y_k$$

where H^u and H^y are lower-triangular Toeplitz matrices of rank N whose elements are the Markov parameters of the system

A Lifting Approach to DRPs -3

$$H^u = \begin{bmatrix} h_0^u & 0 & \cdots & 0 \\ h_1^u & h_0^u & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ h_{N-1}^u & h_{N-2}^u & \cdots & h_0^u \end{bmatrix}$$

$$H^y = \begin{bmatrix} h_0^y & 0 & \cdots & 0 \\ h_1^y & h_0^y & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ h_{N-1}^y & h_{N-2}^y & \cdots & h_0^y \end{bmatrix}$$

$$h_0^u = D, h_0^y = D_0 \text{ and for } i \geq 1, h_i^u = CA^{i-1}B, h_i^y = CA^{i-1}B_0$$

Complex (Iteration) Variable Representation

- Let w be a complex variable and for each $t \in [0, N]$, let the (one-sided) w -transform $W(\cdot)$ be defined as

$$W(\{u_k(t)\}) \doteq \sum_{k=0}^{\infty} u_k(t) w^{-k}$$

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$$\lim_{k \rightarrow \infty} X_k = \lim_{w \rightarrow 1} (w - 1) X(w)$$

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- w -transform is the “z-operator” in the iteration domain.

Operator Perspective of Basic ILC Problem

- For our “plant,” $Y_k = HU_k$, applying the w -transform results in $Y(w) = HU(w)$, a MIMO, constant plant, where $U(w)$ and $Y(w)$ are the w -transforms of U_k and Y_k , respectively



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- This can be written as

$$E(w) = C(w)U(w)$$

where

$$C(w) = \frac{1}{(w-1)}L$$



ILC as a MIMO Control System

- The term

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is effectively the controller of the system (in the repetition domain). This can be depicted as:

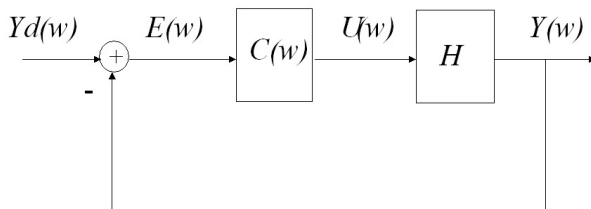


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- Compare this to ILC: $Y(w) = HU(w)$ (thus $H = H_u$).

Higher-Order ILC in the Iteration Domain

- We can use these ideas to develop more general expressions ILC algorithms
- For example, a “higher-order” ILC algorithm could have the form:

$$u_{k+1}(t) = k_1 u_k(t) + k_2 u_{k-1}(t) + \gamma e_k(t+1)$$

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- Next we show how to extend these notions to develop an algebraic (matrix fraction) description of the ILC problem

Closed-Loop Control of DRPs -1

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Closed-Loop Control of DRPs -2

- To proceed, perform lifting to get

$$\begin{aligned} U_{k+1} &= -D_{n-1}U_k - \cdots - D_0U_{k-n+1} \\ &+ N_mE_{k+1} + N_{m-1}E_k + \cdots + N_0E_{k-m+1} \end{aligned}$$



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- Then apply the w -transform and collect terms to get

$$D_c(w)U(w) = N_c(w)E(w),$$

where

$$\begin{aligned} D_c(w) &= w^n I + D_{n-1}w^{n-1} + \cdots + D_1w + D_0 \\ N_c(w) &= N_mw^m + N_{m-1}w^{m-1} + \cdots + N_1w + N_0 \end{aligned}$$

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where

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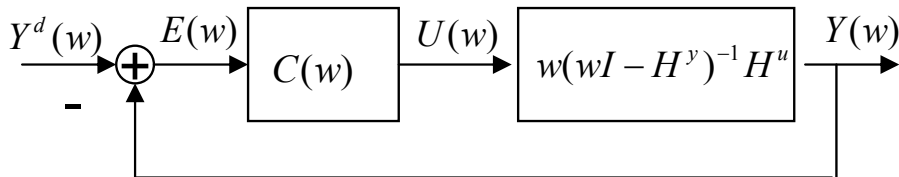
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- The plant and controller combination can be shown as



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- There are two primary concerns: stability and performance, the latter often typified by a tracking requirement
- For both ILC and DRP stability is equivalent to convergence along the iteration axis (as $k \rightarrow \infty$)
- This problem reduces to making the roots of the pole polynomial $G_{cl}(w)$ lie in the open unit disk of the complex plane, where

$$G_{cl}(w) = wH^u[D_c(w)(wI - H^y) + N_c(w)wH^u]^{-1}N_c(w)$$

Complete Framework -1

- In ILC it is common to
 - Separate out current cycle control action, factoring the controller as

$$C(w) = \bar{C}_{ILC}(w) + C_{CITE}$$

where CITE stands for “Current Iteration”



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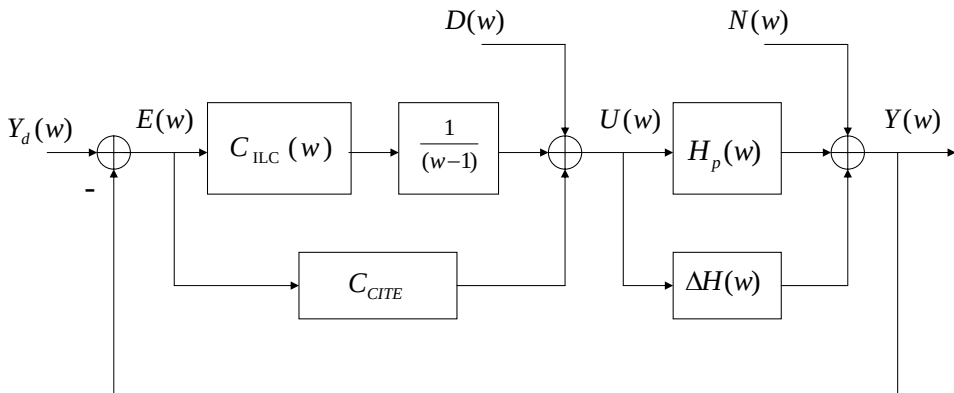
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- Consider the possibility of noise, disturbances, and different types of uncertainty in the plant



Complete Framework -2



Categorization of Problems

Y_d	D	N	C	H	
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	H_p	Classical Arimoto algorithm
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	$H_p(w)$	Rogers'/Owens' multipass problem
$Y_d(z)$	$D(w)$	0	$C(w)$	H_p	General (higher-order) problem
$Y_d(w)$	$D(w)$	0	$C(w)$	H_p	General (higher-order) problem
$Y_d(w)$	$D(w)$	$N(w)$	$C(w)$	$H_p(w) + \Delta(w)$	Most general problem
$Y_d(z)$	$D(z)$	0	$C(w)$	$H(w) + \Delta(w)$	Frequency-domain uncertainty
$Y_d(z)$	0	$w(t), v(t)$	$\Gamma(w-1)^{-1}$	H_p	Least quadratic ILC
$Y_d(z)$	0	$w(t), v(t)$	$C(w)$	H_p	Stochastic ILC (general)
$Y_d(z)$	0	0	$\Gamma(w-1)^{-1}$	H^I	Interval ILC
$Y_d(z)$	$D(z)$	$w(t), v(t)$	$C(w)$	$H(z) + \Delta H(z)$	Time-domain H_∞ problem
$Y_d(z)$	$D(w)$	$w(k, t), v(k, t)$	$C(w)$	$H(w) + \Delta H(w)$	Iteration-domain H_∞ ILC
$Y_d(z)$	0	$w(k, t), v(k, t)$	$\Gamma(k)$	$H_p + \Delta H(k)$	Iteration-varying uncertainty and control
$Y_d(z)$	0	$\tilde{H}$	Γ	H_p	Intermittent measurement problem
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Summary

- ILC, RC, and DRP are classes of **multi-pass systems** that can use data to improve their operation from pass-to-pass
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- By adopting an **operator transform** on the iteration domain (for fixed time), a general **MIMO framework** can be developed for control design problems
- In **Part 2** we consider several **design** problems for these systems, using this framework

Further reading on the algebraic framework

- “A Matrix-Fraction Approach to Higher-Order Iterative Learning Control: 2-D Dynamics through Repetition-Domain Filtering,” Kevin L. Moore, in *Proceedings of 2nd International Workshop on Multidimensional (nD) Systems*, pp. 99-104, Lower Selesia, Poland, June 2000
- “An Algebraic Approach to Iterative Learning Control,” Jari Hatonen, David H. Owens, and Kevin L. Moore, *International Journal of Control*, vol. 77, no. 10, pp. 45-54, January 2004 (also in *Proceedings of the 17th IEEE International Symposium on Intelligent Control*, IEEE ISIC'02, Vancouver, British Columbia October 27-30, 2002)
- “Analysis of Linear Iterative Learning Scheme using Repetitive Process Theory,” D. Owens, E. Rogers, and Kevin L. Moore, *Asian Journal of Control*, vol. 4, No. 1, Mar. 2002, pp. 90-98
- *Iterative Learning Control: An Optimization Paradigm*, David H. Owens, Advances in Industrial Control, Springer-Verlag, London, 2016.
- *Iterative Learning: Control Algorithms and Experimental Benchmarking*, Eric Rogers, Bing Chu, Christopher Freeman, and Paul Lewin, Wiley, 2023